

17. Alice wants to send a large confidential file to Bob, and she wants only Bob to be able to decrypt it. Which of the following hybrid encryption procedures is the most appropriate?

- (A) Randomly generate a session key, encrypt the file using symmetric encryption, and then encrypt the session key using Bob's public key
- (B) Encrypt the file using Alice's public key, and then send Alice's private key to Bob
- (C) Directly encrypt the entire file using Bob's private key
- (D) Use a hash function to encrypt the file, and then send the hash value to Bob
- (E) Use Bob's public key to generate a digital signature, and then use the signature to decrypt the file

18. An internal host 192.168.1.10:5000 connects to an external server address and port 203.0.113.8:40001. The NAT device maps this connection to the public the server, which source and destination are the most reasonable?

- (A) source = 192.168.1.10:5000, destination = 198.51.100.20:443
- (B) source = 198.51.100.20:443, destination = 203.0.113.8:40001
- (C) source = 203.0.113.8:443, destination = 192.168.1.10:5000
- (D) source = 192.168.1.10:40001, destination = 203.0.113.8:443
- (E) source = 203.0.113.8:40001, destination = 198.51.100.20:443

19. Which of the following statements about the complexity classes P and NP is correct?

- (A) NP stands for "non-polynomial," meaning such problems cannot be solved in polynomial time
- (B) Every problem in P is also in NP
- (C) NP-complete problems cannot be solved by any algorithm
- (D) If a problem is in NP, solving it must take exponential time
- (E) P and NP are disjoint sets of problems

20. For which of the following applications is UDP generally more suitable than TCP?

- (A) Downloading a software update, where the file must arrive completely intact
- (B) Sending an e-mail with attachments
- (C) On-line banking transactions
- (D) Transferring a file using FTP
- (E) Real-time live video streaming, where low latency matters and occasional packet loss is acceptable